Customer Segmentation Report

1. Introduction

Customer segmentation is a crucial process for businesses to categorize their customers based on similar characteristics and behaviors. This report outlines the customer segmentation process using clustering techniques on customer profile data (Customers.csv) and transaction history (Transactions.csv). The goal is to derive meaningful customer clusters and evaluate their effectiveness using the Davies-Bouldin (DB) Index.

2. Clustering Approach

2.1 Data Sources

- **Customers.csv**: Contains demographic and profile-related details of customers.
- **Transactions.csv**: Contains transactional behavior of customers.

2.2 Methodology

- Data preprocessing including handling missing values, normalization, and feature selection.
- Applied **Principal Component Analysis (PCA)** to reduce dimensionality and improve visualization.
- Used a clustering algorithm to segment customers (final selection based on evaluation metrics).
- Evaluated clustering performance using **DB Index** and **Silhouette Score**.
- Visualized the clusters to interpret segmentation results.

3. Clustering Results

3.1 Number of Clusters

- The optimal number of clusters selected was **4**, based on evaluation metrics and business interpretability.

3.2 Clustering Metrics

- **Davies-Bouldin Index (DB Index)**: **1.171**
    - A lower DB Index value indicates better clustering quality.
- **Silhouette Score**: **0.336**
    - A higher silhouette score (closer to 1) indicates better-defined clusters.

3.3 Cluster Visualization

- PCA was used to visualize clusters in two dimensions.
- The scatter plot (attached) illustrates the formed clusters with different colors, showing separation between them.

4. Insights & Interpretation

- The clustering approach successfully segmented customers into four distinct groups.
- The DB Index of **1.171** suggests a reasonably good cluster separation and compactness.
- The **Silhouette Score of 0.336** indicates moderate clustering performance, suggesting room for improvement.
- The visualization confirms meaningful differentiation among customer groups.

5. Recommendations & Future Work

- Further tuning of the clustering algorithm (e.g., trying different distance metrics or clustering techniques) could improve segmentation quality.
- Incorporating additional features such as customer sentiment or online behavior may enhance clustering effectiveness.
- Exploring deep learning-based clustering approaches could yield better results.

6. Conclusion

This report outlines an effective approach to customer segmentation using clustering techniques. The analysis identified **four customer clusters**, with a DB Index of **1.171**, indicating a well-structured clustering model. The insights derived can help businesses tailor marketing strategies, improve customer engagement, and enhance business decisions.